

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460205.6655 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1759 +/- 0.0097
Transit depth (Rp/Rs)^2	3.09 +/- 0.34 [%]
Orbital Inclination (inc)	88.16 +/- 0.45
Airmass coefficient 1 (a1)	0.9418 +/- 0.0092
Airmass coefficient 2 (a2)	0.047 +/- 0.033
Scatter in the residuals of the lightcurve fit is	0.97 %
Best Comparison Star	#1 - [537, 207]
Optimal Aperture	5.13
Optimal Annulus	7.96
Transit Duration (day)	0.0848 +/- 0.0032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1759 $\pm$ 0.0097 vs 0.1714 (deviation 0.43 σ)
Duration fit vs published	0.0848 d vs 0.0892 d (deviation 1.26 σ)
Mid-time O-C	5.4 min
Depth SNR	16.7
Residual scatter	0.97 %

Light curve

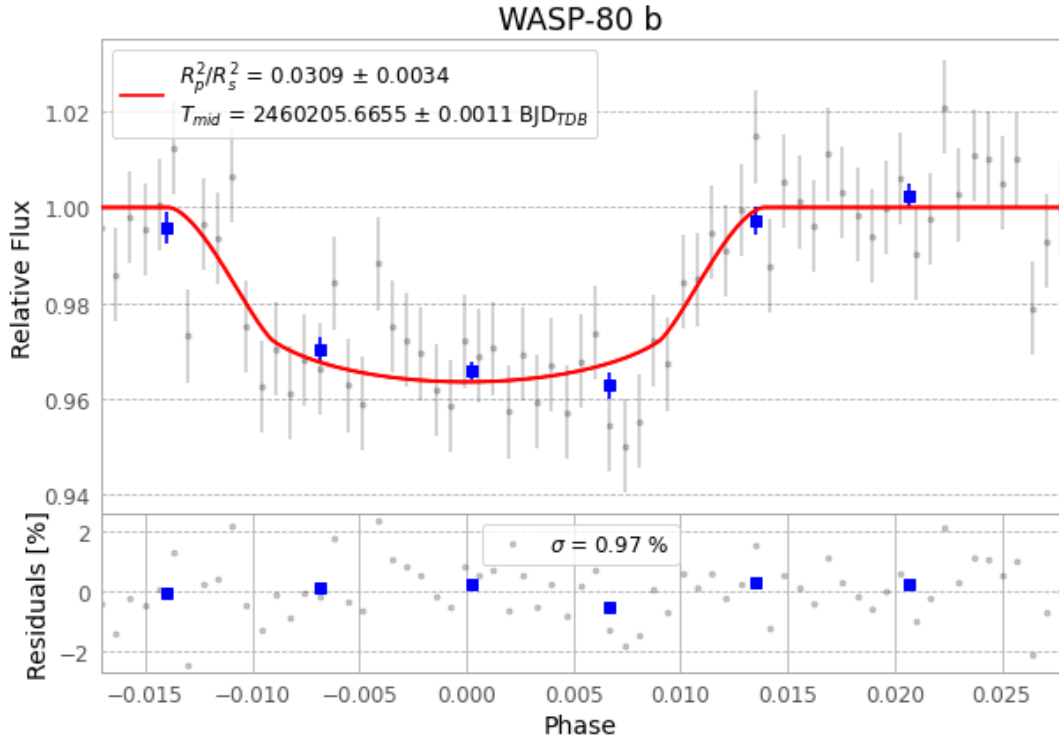


Figure 1 — FinalLightCurve_WASP-80 b_2023-09-17.png (SHA-256 7dd1b256c97ee2a6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [305, 188], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cc8310eaaef9920a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

67 FITS frames (44 MB), WASP-80230918023615.FITS → WASP-80230918055415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-80-230918.log (37a40bee3bbf...), FinalLightCurve_WASP-80 b_2023-09-17.png (7dd1b256c97e...), FinalLightCurve_WASP-80 b_2023-09-17.csv (1d322c163e6e...)

Compute resources

Wall time 122.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 23:30Z.